free doesn’t mean that our behaviour is not determined by the laws of nature. It means that it is determined by the laws of nature acting in our brains.

Our free decisions are freely determined by the results of the rich and fleeting interactions between the billion neurons in our brain: they are free to the extent that the interaction of these neurons allows and determines. Does this mean that when I make a decision it’s ‘I’ who decides? Yes, of course, because it would be absurd to ask whether ‘I’ can do something different from what the whole complex of my neurons has decided: the two things, as the Dutch philosopher Baruch Spinoza understood with marvellous lucidity in the seventeenth century, are the same.

There is not an ‘I’ and ‘the neurons in my brain’. They are the same thing. An individual is a process: complex, tightly integrated.

When we say that human behaviour is unpredictable, we are right, because it is too complex to be predicted, especially by ourselves. Our intense sensation of internal liberty, as Spinoza acutely saw, comes from the fact that the ideas and images which we have of ourselves are much cruder and sketchier than the detailed complexity of what is happening within us. We are the source of amazement in our own eyes.

We have a hundred billion neurons in our brains, as many as there are stars in a galaxy, with an even more astronomical number of links and potential combinations through which they can interact. We are not conscious of all of this. ‘We’ are the process formed by this entire intricacy, not just by the little of it of which we are conscious.

The ‘I’ who decides is that same ‘I’ which is formed (in a way that is still certainly not completely clear, but which we have begun to glimpse) from reflections upon itself; through self-representations in the world; from understanding itself as a variable point of view placed in the context of the world; from that impressive structure that processes information and constructs representations which is our brain. When we have the feeling that ‘it is I’ who decides we couldn’t be more correct. Who else?

I am, as Spinoza maintained, my body and what happens in my brain and heart, with their immense and, for me, inextricable complexity.

The scientific picture of the world which I have related in these pages is not, then, at odds with our sense of ourselves. It is not at odds with our thinking in moral and psychological terms, or with our emotions and feelings. The world is complex, and we capture it with different languages, each appropriate to the process which we are describing. Every complex process